ALICE L WALTON SCHOOL OF MEDICINE

Diagnosis of Heart Failure: From Clinical Presentation to Classification

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Adapted from: Diagnosis of Heart Failure

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0:00 / 37:10 1x

Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Explain the clinical and morphologic features of left-sided heart failure (HF) and list the most common causative disease.
2. Explain the clinical and morphologic features of right-sided heart failure and l[illegible]
3. Describe how echocardiogram, serum studies, and radiology are used for HF diagnosis
4. Differentiate diagnosis of heart failure from conditions that present with symptoms similar to heart failure
5. Describe how to quantify the severity and functional status of a patient with heart failure using the New York Heart Association (NYHA) system.

CASE CONNECTION

CL is a 56-year-old male who comes to his physician with shortness of breath, which has increased during the last few weeks, during both rest and exertion. He has diabetes mellitus, well controlled with metformin. He has no chest pain, but he says the shortness of breath awakens him at night. His cardiac exam is normal, and he has a few crackles at both lung bases. His pulse, blood pressure, and respiratory rates are normal, and he shows trace pedal edema. His ECG is only abnormal for diffuse inverted T waves.

What should be the additional evaluation for CL's shortness of breath? Consider your answer as you read, and we'll revisit CL at the end of the brick.

GO TO CONCLUSION ↓

What Is Heart Failure?

Heart failure (HF), so[illegible]art failure (CHF), is not a single disease. Rather, it is a clinical **syndrome**, a collection of symptoms (reported by the patient) and signs (seen by the clinician). Rather than being a single disease entity, heart failure represents the final common pathway of numerous cardiac conditions that compromise the heart's ability to fill with or eject blood effectively. Heart failure can be caused by numerous cardiac diseases or conditions such as right ventricular dysfunction, left ventricular (LV) dysfunction, pericardial diseases, valvular disease, obstructive lesions in the heart or great vessels and high output heart failure.

All forms of heart failure (HF) involve the heart's inability to pump enough blood to meet the body's needs. This can happen when weakened ventricles cannot contract effectively, leading to systolic HF, or Heart Failure with reduced Ejection Fraction **(HFrEF)**.

HF may also occur when the ventricles become stiff due to conditions such as hypertension, aortic stenosis, restrictive or infiltrative diseases, or hypertrophic cardiomyopathy. This reduced compliance causes diastolic dysfunction and results in Heart Failure with preserved Ejection Fraction **(HFpEF)**.

This distinction is crucial because it guides both diagnostic approach and therapeutic management.

HF can also be divided into right- and left-sided, based on symptoms and signs, although in most patients there is a combination of both left and right and the most common cause of right heart failure, is left heart failure.

Systolic vs Diastolic HF

Systolic heart failure aka (HFrEF), is characterized by impaired ventricular contractility resulting in reduced cardiac output. The left ventricular ejection fraction (LVEF) is typically less than 40%. The ventricle dilates and remodels, adopting a more spherical shape that further compromises efficient contraction. This form of HF is often associated with coronary artery disease, previous myocardial infarction, or dilated cardiomyopathy

Diastolic heart failure aka (HFpEF) occurs when the ventricle cannot relax or fill properly during diastole despite normal contractile function. The LVEF is preserved at greater than 50%. The ventricular walls are often thickened and stiff, limiting diastolic filling and increasing filling pressures. Common causes include hypertension, aging, diabetes, and hypertrophic cardiomyopathy. Early data is now showing chronic inflammation may also cause HFpEF.

CLINICAL CORRELATION

A third category, Heart Failure with mid-range Ejection Fraction **(HFmrEF)**, describes patients with an LVEF between 40-49% who share features of both HFrEF and HFpEF.

The distinction between these types has significant clinical implications. HFrEF has well-established guideline-directed medical therapies that improve outcomes, while effective treatments specifically for HFpEF remain more elusive.

Additionally, as mentioned in a prior brick, the auscultatory findings differ: systolic dysfunction [illegible]stolic dysfunction manifests an S4 gallop.

Clinical and Morphologic Features (LO #1)

The clinical and morphologic features of HF reflect both structural and functional cardiac abnormalities. These changes alter cardiac hemodynamics, leading to increased filling pressures, reduced cardiac output, and neurohormonal activation that further exacerbates the condition.

In HFrEF, the ventricle becomes dilated and thin-walled, increasing end-diastolic volume and reducing contractile efficiency. The chamber becomes more spherical, weakening fiber mechanics. In HFpEF, the ventricle develops concentric hypertrophy with a smaller or normal chamber size. The thickened, stiff wall limits relaxation and impairs diastolic filling.

What Signs and Symptoms Distinguish Left-Sided Versus Right-Sided Heart Failure?

We can divide most of the findings of left-sided HF into "forward" and "backward" signs and symptoms. When the LV doesn't function properly, blood does not travel forward into the circulation at an adequate rate, depriving the organs of oxygen (decreased cardiac output) but, in addition, the heart cannot empty properly, so blood builds up behind the failed ventricles in the lungs and/or the venous system. This is "backward" failure. We will next describe some of the findings of each of these types of HF.

Left-Sided HF Findings: Effects of Decreased Organ Perfusion

In patients with seve[illegible]lood flow to the body, leading to chara[illegible] findings include:

- **Tachycardia:** Low BP sensed at the carotid and aortic baroreceptors activates the sympathetic nervous system, increasing heart rate; ventricular stretch from volume overload further augments this response.
- **Mental status changes:** Confusion, dizziness, or loss of consciousness result from decreased cerebral perfusion.
- **Edema:** Peripheral edema develops when reduced cardiac output lowers renal perfusion, triggering RAAS and sympathetic activation. This increases renal sodium and water retention, expands plasma volume, and leads to edema.
- **Kidney injury:** Chronic reductions in renal perfusion and elevated venous pressure can lower renal blood flow and GFR, often leading to increased creatinine
- **Pulmonary crackles** (aka rales): Fine, crackling sounds heard on auscultation, typically at the lung bases
- **S3 gallop**
- **S4 gallop**
- **Laterally displaced point of maximal impulse (PMI):** Indicates LV enlargement

The cardinal symptoms of left-sided heart failure include:

- **Dyspnea on exertion (DOE):** Often the earliest symptom, reflecting increased pulmonary pressures during physical activity
- **Orthopnea:** Shortness of breath when lying flat (supine), typically relieved by using multiple pillows or sitting upright. Due to fluid shifting to the lungs when lying down, worsening pulmonary congestion.
- **Paroxysmal nocturnal dyspnea (PND):** Episodes of shortness of breath that wake the pa[illegible]
- **Fatigue & weakness:** decreased cardiac output and reduced oxygen delivery to skeletal muscles
- **Decreased exercise tolerance:** Progressive limitation of physical activity

Cough and DOE are common but nonspecific for heart failure. **Orthopnea and paroxysmal nocturnal dyspnea (PND) are highly specific**, indicating a cardiac rather than pulmonary cause.

Signs and symptoms found in left-sided HF are shown in Figure 1.

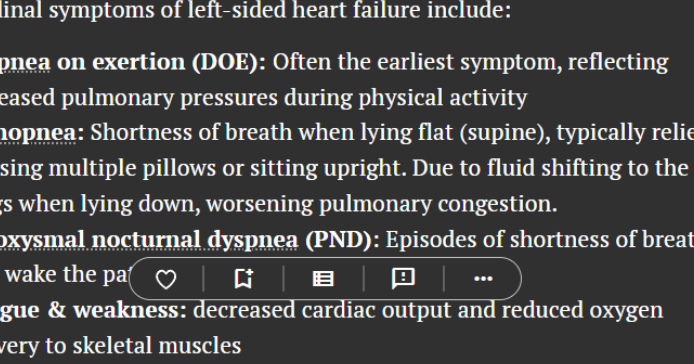

Figure 1 Signs and symptoms of left-sided heart failure

Why do patients with systolic HF have tachycardia?

What Are Signs and Symptoms of Right-Sided Heart Failure? (LO #2)

Effects of "Backed-Up" Blood

In right-sided HF, impaired forward flow into the pulmonary circulation causes blood to back up from the right atrium into systemic veins, leading to congestion in any vessels draining into the right atrium. This can result in:

- **Jugular venous distention (JVD):** This occurs when blood backs up into the superior vena cava and then the jugular veins.
- **Liver congestion:** Right-sided HF patients may have jaundice, hepatomegaly (liver enlargement), and a positive hepatojugular reflux on exam, a syndrome termed **"congestive hepatopathy."** In pathology often called a "nutmeg[illegible]
- **Ascites:** When ve[illegible] out of the vessels into the peritoneal cavity of the abdomen.
- **Reduced GFR:** Elevated venous pressure raises post-glomerular pressure and lowers GFR, contributing to chronically higher creatinine in HF, activating RAAS, further worsening fluid retention.
- **Peripheral edema:** Edema, typically bilaterally and manifests as a pitting type of edema that is **gravity dependent.**

CLINICAL REASONING

A positive hepatojugular reflex occurs on exam when there is increased jugular venous pressure with abdominal compression.

Additional findings may include abdominal discomfort from hepatic congestion, anorexia, and early satiety due to visceral congestion, and weight gain from fluid retention. In severe cases, jaundice may develop from congestive hepatopathy. Right-sided heart failure rarely causes significant dyspnea unless left-sided failure is also present or there is a large pleural effusion.

Signs and symptoms found in right-sided HF are shown in Figure 2.

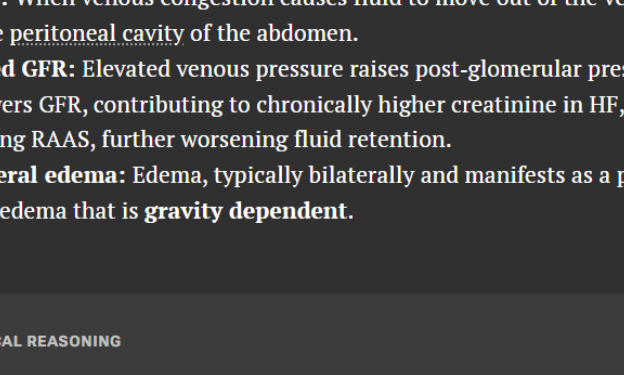

Figure 2 Signs and symptoms of right-sided heart failure

Mixture of Signs and Symptoms

While it is useful to characterize the signs and symptoms of HF into left- and right-sided categories, patients will often present with a mixture of the two. As heart failure progresses, most patients develop biventricular failure with mixed clinical features. Also, patients may have some but not all of these findings. For example, not every patient with left-sided HF will have pulmonary edema.

Some patients will only present with lower extremity edema or isolated shortness of breath. The clinician's job is to look for signs and symptoms of HF and test further if needed.

Mnemonic

How does HF lead to peripheral edema?

What Tests Are Used to Diagnose Heart Failure? (LO #3)

The best test to **confirm HF** is an **echocardiogram** ("**echo**"), but other tests may also be useful. Diagnosing requires a systematic approach that integrates clinical findings with appropriate diagnostic testing. While the history and exam provide crucial information, objective testing is necessary to confirm the diagnosis, determine the type of heart failure (HFrEF, HFmEF, HFpEF), identify underlying causes, and assess severity. A multimodal diagnostic strategy typically includes imaging, labs, and sometimes more specialized testing.

Electrocardiogram

The ECG is often the first test done for suspected cardiac disease and should be done in all patients with HF. Unfortunately, it has limited utility for HF diagnosis, meaning it is not diagnostic of HF on its own. The ECG may reveal prior myocardial infa[illegible]ertrophy (LVH), Left Bundle Branch Block (LBBB, common in HFrEF), Atrial fibrillation, other arrythmias, nonspecific ST-segment and T-wave abnormalities.

For HFpEF patients with LVH, most will show high voltage in the lateral ECG leads (eg, I, V_5-V_6) due to the high left ventricular (LV) mass, with a corresponding high electrical signal. They may also show a "strain pattern" of ST depressions and T-wave inversions in the lateral leads (Figure 3).

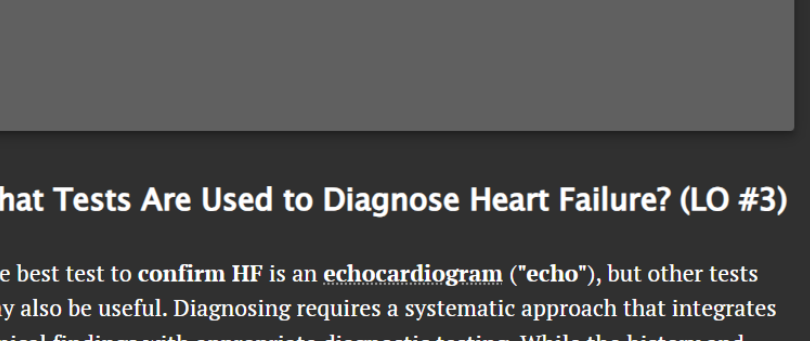

Figure 3 Left ventricular hypertrophy on ECG

CLINICAL CORREL[illegible]

A common criterion for high voltage is to add the S amplitude in V_1 to the R amplitude in V_5 or V_6. If >35 mm, there is LVH. Here the calculation shows V_1 (20 mm) + V_5 (25 mm) = 45, indicating LVH.

ECGs may also detect an acute or prior MI or arrhythmia that may be causing HF symptoms.

Imaging

Transthoracic Echocardiography (TTE). This is the **best test** for diagnosing HF and **should be done in all patients with suspected HF.** It can estimate the LVEF, diagnose abnormalities of diastolic filling, determine the ventricular wall thickness, assess for valve disease such as stenosis or regurgitation that can cause or exacerbate HF, and detect areas of abnormal wall motion seen from past myocardial infarctions. It can also give an estimation of pulmonary artery pressures to assess for pulmonary hypertension.

It distinguishes between HFrEF (EF <40%), HFmrEF (EF 40-49%), and HFpEF (EF ≥50%), which have different treatment approaches.

Chest X-Ray-This will often detect an enlarged heart (cardiomegaly) but this finding is seen in both systolic and diastolic HF so is not useful in determining the cause of HF. While CXR findings lack specificity for HF, they provide valuable info about pulmonary congestion and can help rule out alternative diagnoses such as pneumonia. Notably, a normal CXR **does not** exclude heart failure, particularly in well-compensated or early disease. CXRs can detect pulmonary vascular [illegible]

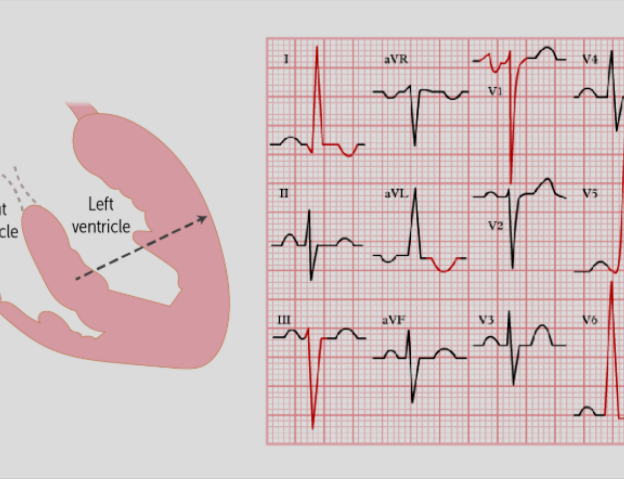

Figure 4 Pulmonary edema in a patient with left heart failure due to myocardial infarction

Coronary Angiography. This is not done routinely for the diagnosis of HF but should be done when there is ECG or lab evidence of myocardial infarction in the setting of acute HF symptoms indicating ischemic heart disease.

Cardiac MRI-Provides detailed tissue characterization, useful for assessing cardiomyopathies, myocarditis, or infiltrative diseases such as amyloidosis.

Nuclear imaging: May assess myocardial viability or perfusion defects in ischemic cardiomyop[illegible]

CT coronary angiography: Non-invasive alternative to assess coronary anatomy in selected patients

Serum Studies– Labs

Natriuretic peptides (atrial and brain) —primarily B-type natriuretic peptide (BNP) and N-terminal pro-BNP (NT-proBNP)—are biomarkers released by ventricular myocytes in response to increased wall stress. These peptides serve as valuable diagnostic tools, particularly in settings where the clinical presentation is ambiguous. They then cause increased renal sodium excretion.

The test is sensitive, but **not specific** for HF, so a positive test must be confirmed with another more precise test such as an echo. BNP is best used when **ruling out HF** since a negative test is convincing evidence that no HF is present which means the test has a **high negative predictive value.** False positive tests are seen in patients with CKD, older patients, and females. Moderate positive predictive value (PPV) means that elevated levels support but do not confirm HF. Additionally, BNP levels tend to be lower in obese individuals, meaning a negative can be a false negative.

Patients with new or acutely worsened symptoms of HF should always have **serum troponin levels** drawn to rule out acute myocardial infarction. Cardiac troponins may be elevated with underlying ischemia. A **CBC** can identify anemia which can exacerbate or mimic HF, or infection. **Renal function tests** assess kidney function, which affects both prognosis and medication choices. **Liver function tests** may be elevated due to hepatic congestion in right-sided heart failure. **Thyroid function tests** can evaluate for thyroid disorders can cause or worsen heart failure.

Mnemonic

Why and how is the brain natriuretic peptide (BNP) level useful in HF diagnosis?

How Do We Apply Diagnostic Criteria and Clinical Findings to Differentiate Heart Failure From Mimics? (LO #4)

HF is often confused with other diagnoses, since symptoms like edema and dyspnea have a variety of causes. HF should be considered when any patient has acute dyspnea.

Pneumonia frequently presents with dyspnea, cough, and occasionally chest pain, mimicking heart failure. Key distinguishing features include fever, leukocytosis, productive cough with purulent sputum, localized crackles or bronchial breath sounds, CXR showing focal consolidation (although pneumonia can be bilateral).

Pulmonary embolism (PE) can mimic heart failure with acute dyspnea, chest pain, and occasionall[illegible] include **sudden onset** of symptoms, pleuritic chest pain and hemoptysis, risk factors such as immobilization, recent surgery, malignancy or thrombophilia, elevated D-dimer (though not specific), CT pulmonary angiography showing filling defects.

Chronic obstructive pulmonary disease (COPD) and asthma exacerbations present with wheezing and dyspnea that can be confused with HF. Distinguishing characteristics include prominent and diffuse wheezing, history of allergic triggers or respiratory infections, lack of orthopnea and PND, response to bronchodilators and of course pulmonary function tests showing an obstructive pattern.

Acute respiratory distress syndrome (ARDS) also sometimes referred to as noncardiogenic pulmonary edema presents with severe hypoxemia and diffuse pulmonary infiltrates that can resemble cardiogenic pulmonary edema. Key differences include identifiable precipitating factor (sepsis, trauma, aspiration), normal cardiac function on echo. A higher pulmonary artery occlusion pressure (>18 mmHg) is often present in cardiogenic edema from HF. The CXR pattern may be more heterogeneous in ARDS compared to the more symmetric pattern in HF but truthfully, look quite similar.

Table 1 shows how to differentiate HF from other common conditions causing acute dyspnea.

Table 1 Differential diagnosis of acute dyspnea

Disease	Supports HF	Supports other disease	Notes
Pneumonia	[illegible]	[illegible]	Chest x-ray can look similar to HF
Pulmonary embolus	Abnormal chest x-ray Peripheral edema	Calf tenderness Pleuritic chest pain +D dimer	Physical exercise may cause dyspnea. Brain natriuretic peptide (BNP), troponins may be elevated in either
Asthma	Abnormal chest x-ray	Wheezing Allergic symptoms	Wheezing may be present in congestive heart failure ("cardiac asthma")
Noncardiogenic pulmonary edema (ARDS)	Echocardiogram	Fever Hypotension Serious infection	Chest x-rays will look very similar

Formulating a Heart Failure Diagnosis

Due to the variable manifestations of heart failure, making the diagnosis can be very tricky since other diseases can have overlapping findings (Table 1). Some clinical signs and symptoms are much more predictive than others. Here are some tips.

These findings are **very specific for HF**, but have **limited sensitivity** (not seen in all patients). In other words, have a high PPV.

- **Symptoms:** orthopnea, PND
- **Signs:** JVD, S_3 heart sounds, laterally displaced point of maximal impulse (PMI, due to cardiac enlargement)
- Echocardiography: The gold standard for confirming heart failure diagnosis

Diagnostic tests with[illegible]

- Echo: The gold standard for confirming HF diagnosis and critical for classifying the type: HFrEF, HFmEF, HFrEF
- BNP <100 pg/mL or NT-proBNP <300 pg/mL: Effectively rules out acute HF
- Chest X-ray showing cardiomegaly, pulmonary vascular redistribution, and interstitial edema: Strongly supports HF diagnosis
- Response to diuretic therapy: Rapid improvement with diuresis suggests fluid overload consistent with HF

Other Signs and Symptoms

Patients with less specific findings (eg, dyspnea, edema, fatigue, cardiomegaly on chest x-ray) should receive BNP (so HF can be excluded) and/or echocardiography (to confirm and classify the HF).

What other conditions can have symptoms similar to heart failure?

How Do We Classify the Severity of Heart Failure? (LO #5)

There are ways that all these signs and symptoms can be quantified among HF patients for clinical and research purposes. We will review the **New York Heart Association (NYHA) classification** here.

The NYHA grades HF based on frequency of **clinical symptoms** during activity

The physical exam and labs are not included because frequently the degree of disability does not correlate well with any single diagnostic parameter. **Class I** is mild, and **Class IV** is the most severe.

- NYHA Class I patients have no symptoms and may not receive treatment. They would include patients who have a diagnosed cardiomyopathy on lab testing (like a LVH on ECG) but who remain asymptomatic. These patients often have compensated HF and may only be identified through screening or evaluation for other conditions.
- NYHA Class II patients experience mild symptoms (dyspnea, fatigue, palpitations) during ordinary activities but are comfortable at rest. They might limit certain activities but maintain relatively normal daily function. For example, a Class II patient might experience DOE when climbing more than one flight of stairs or walking uphill but can perform most household tasks without symptoms. Considered mild HF
- NYHA Class III indicates moderate-to-severe HF with significant functional limitations. These patients are comfortable only at rest and experience symptoms with minimal exertion, such as walking short distances on level ground or performing light housework. They typically require frequent adjustment of medical therapy and closer monitoring Considered mod[illegible]
- NYHA Class IV represents the most advanced and severe stage of HF, where patients experience symptoms even at rest. These patients are gravely disabled. Any activity increases discomfort, and these patients are often unable to perform activities of daily living independently. Class IV patients frequently require hospitalization, advanced therapies (such as inotropic support), or consideration for mechanical circulatory support or transplantation.

Mnemonic

What is the basis of the New York Heart Association (NYHA) scale?

Clinical Applications

The NYHA classification system serves several important clinical purposes:

- **Prognostic assessment:** Higher NYHA class correlates with increased mortality and hospitalization risk
- **Treatment guid[illegible]** indicated for higher NYHA classes
- **Monitoring disease progression or improvement:** Changes in NYHA class reflect response to therapy
- **Communication among healthcare providers:** Provides standardized terminology for describing functional status
- **Research stratification:** Allows comparison of patient populations across clinical trials

While the NYHA classification is valuable, it has limitations. The assessment is subjective and may vary between observers or over time. Additionally, comorbidities like obesity, lung disease, or peripheral vascular disease can cause similar symptoms, potentially confounding the classification. To address these limitations, objective measures of functional capacity, such as the 6-minute walk test or cardiopulmonary exercise testing, may provide complementary information.

The NYHA classification also informs treatment decisions. For example:

- NYHA Class I: Focus on preventing disease progression with neurohormonal antagonists (ACE inhibitors, beta-blockers)
- NYHA Class II: Addition of mineralocorticoid receptor antagonists and potentially SGLT2 inhibitors
- NYHA Class III: Consider device therapy (ICD, CRT) and more intensive diuretic regimens
- NYHA Class IV: Evaluation for advanced therapies including inotropes, mechanical circulatory support, or transplantation

It's important to note that NYHA class may fluctuate over time, reflecting both the natural history of heart failure and response to interventions. Patients may improve from Class I[illegible]r deteriorate from Class II to IV during acute decompensation. Regular reassessment of NYHA class provides valuable information about disease trajectory and treatment efficacy.

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to CL, what would be the next step in his diagnosis?

CL has several findings that are specific for heart failure, specifically paroxysmal nocturnal dyspnea (PND). He has a risk factor for ischemic heart disease (and HF), ie, diabetes mellitus. However, he does not have abnormalities on his cardiac exam, and his nonspecific T wave inversions are not specific for HF.

Given the uncertainty of his HF diagnosis, a brain natriuretic peptide (BNP) level would be reasonable. If normal, we can rule out HF, and CL should proceed with more thorough pulmonary testing. If the BNP is high, he should receive echocardiography, the best test to diagnose and confirm HF.

CL had an elevated BNP, and his echo showed HF with a decreased ejection fraction of 28%. He was begun on appropriate medications and his PND immediately improved. He is breathing and sleeping better now.

Summary

- Heart failure (HF) is a **clinical syndrome** resulting from various cardiac conditions that impair the heart's ability to fill with or eject blood effectively.
- HF is categorized[illegible]d diastolic heart failure (HFpEF) each with distinct pathophysiological mechanisms and treatment approaches.
- Left-sided HF is characterized by symptoms such as dyspnea, orthopnea, and PND due to decreased organ perfusion and pulmonary congestion.
- Right-sided HF leads to symptoms like JVD, liver congestion, and edema due to blood backing up into the systemic circulation.
- Echo is the gold standard for diagnosing HF, while BNP levels are useful for ruling out HF in ambiguous cases.
- The New York Heart Association (NYHA) classification grades HF severity from Class I (no symptoms) to Class IV (severe symptoms), guiding prognosis and treatment decisions.

Review Questions

1. What is the most common cause of right-sided heart failure (HF)?

A. ● Left-sided heart failure
B. ○ Mitral regurgitation
C. ○ Myocardial infarction
D. ○ Systemic hypertension
E. ○ Tricuspid stenosis

Hide Explanation

The correct answer is left-sided heart failure (A). Most right-sided heart failure (HF) is caused by the same disorder that caused left-sided HF. Mitral regurgitation (B) and myocardial infarction (C) would cause combined right and left HF, not isolated right HF. Systemic hypertension (D) leads to left-sided HF, not right. Tricuspid stenosis (E) limits blood flow from right atrium to right ventricle and may cause right atrial enlargement, but does not cause right heart failure.

2. Which of the following is a symptom of left-sided HF but not right-sided HF?

A. ○ Ascites
B. ○ Hepatic congestion
C. ○ Jugular venous distension
D. ○ Lower extremity edema
E. ● Pulmonary congestion

Hide Explanation

The correct answer is pulmonary congestion (E), which results from left ventricular dysfunction and congestion of fluid in the pulmonary circulation. All the other answer choices—ascites (A), hepatic congestion (B), jugular venous distension (C), and lower extremity ed[illegible]e, but only when the left-sided failure has [illegible]ese symptoms of congestion are in the systemic circulation and therefore can be seen in both left- and right-sided failure.

3. Which of the following is best characterized by impaired relaxation and impeded ability of the ventricle to fill with blood?

A. ● Diastolic dysfunction
B. ○ Left-sided heart failure
C. ○ Right-sided heart failure
D. ○ Systolic dysfunction

Hide Explanation

The correct answer is diastolic dysfunction (A), which is characterized by impaired relaxation and impeded ventricular filling with blood. Left-sided heart failure (B) and right-sided heart failure (C) are descriptions of clinical syndromes, not mechanisms of ventricular dysfunction. Systolic dysfunction (D) is when the ventricle has decreased force of contraction and difficulty pumping blood.

4. A 55-year-old male presents with fatigue during day-to-day activities. He is still able to work and walk up one flight of stairs, and he is usually without symptoms. He do[illegible]en he walks several blocks at a time, and he noticed some mild chest pain while shoveling snow in his driveway. He is diagnosed with heart failure. His symptoms are most consistent with which New York Heart Association class?

A. ○ I
B. ● II
C. ○ III
D. ○ IV

Hide Explanation

The correct answer is New York Heart Association (NYHA) class II (B). NYHA class II is characterized by mild limitation in physical activity. Ordinary physical activity results in fatigue, palpitation, dyspnea, or angina pectoris (mild congestive HF) in this class. Class I (A) would be asymptomatic. Class III (C) would cause marked limitation in activity, and normal activities would always cause symptoms. Class IV (D) is the worst and shortness of breath and pain at rest can be characteristic

5. Which of the following diagnostic tests would be most diagnostic of left ventricular hypertrophy in a patient with nonsystolic heart failure?

A. ○ Cardiomegaly on chest x-ray
B. ○ Elevated[illegible]
C. ● High-amplitude QRS complexes on lateral leads on ECG
D. ○ PR prolongation on ECG
E. ○ ST elevations on ECG

Hide Explanation

The correct answer is high amplitude QRS complexes on lateral leads on ECG (C). The increased left ventricle (LV) mass increases the electrical signal. Cardiomegaly on chest x-ray (A) can be seen in either systolic or diastolic heart failure (HF). Elevated serum troponins (B) are concerning for myocardial infarction. PR prolongation on ECG (D) suggests first degree atrioventricular (AV) block, not specific for HF. ST elevations on ECG (E) suggest myocardial infarction or pericarditis.

How would you rate this Brick?

👍 👎

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the clinical and morphologic features of left-sided heart failure (HF) and list the most common causative disease.~~
02 ☑ ~~Explain the clinical and morphologic features of right-sided heart failure and list the single most common cause.~~
03 ☑ ~~Describe how echocardiogram, serum studies, and radiology are used for HF diagnosis~~
04 ☑ ~~Differen[illegible]ions that present with sy[illegible]~~

- Kiefer T, Granger CB, Jackson KP. Heart Failure. In: Papadakis MA, McPhee SJ, Rabow MW, McQuaid KR. eds. *Current Medical Diagnosis & Treatment 2023*. McGraw Hill; 2023: e1-22.
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Objective Confidence: 5/5